

Md Nasiat Hasan Fahim

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Education

Shahjalal University of Science and Technology

B.Sc.(ENGG.) IN COMPUTER SCIENCE & ENGINEERING

CGPA: **3.89** / 4.00 · Class Rank: **3rd** of 111

Sylhet, Bangladesh

03/2022 – 07/2026

Experience

BRAC Tech & SELP-RHRN AI Fellowship

BACKEND SOFTWARE ENGINEER

- A specialized AI fellowship to engineer a gamified platform addressing gender discrimination mindsets for BRAC, the world's largest NGO. FastAPI backend for a gamified social platform, Multi-LLM pipelines (GPT-4o, Gemini) for real-time Bengali content like book, research paper web search and suggestions, PostgreSQL (JSONB) schemas, deployment via Docker for containerized microservices.

Dhaka, Bangladesh

07/2025 – 12/2025

Shahjalal University of Science and Technology

RESEARCH ASSISTANT

- Worked under Dr. Ahsan Habib on Crime Pattern Analysis. Analyzed spatiotemporal crime patterns in Sylhet using K-means clustering to identify high-risk temporal hotspots and time series forecasting for future crime prediction with newspaper reporting and police data. Developed 3D visualization tools to map crime clusters and victim-offender relationships.

Sylhet, Bangladesh

01/2025 – 06/2026

Publications

Real-Time Earthquake Magnitude Classification from Initial P-Waves: Models, Dataset, and Comparative Analysis for South Asia

M. N. Hasan Fahim, M. A. Ullah Muhib, R. A. Tanvir, and A. Al Noman. *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 5175–5180. doi: 10.1109/ICCIT68739.2025.11489542.

- Curated 7,318 South Asian earthquake events and benchmarked six models using 7-second single-station P-waves; the Transformer achieved 76.23% accuracy, 81.56% adaptive accuracy, and 4.8 ms inference latency.

Protein-Based Fish Species Identification: Dataset, Models, and Insights from Native Bangladeshi Fish

N. H. Fahim, A. U. Muhib, and M. S. Rahman. *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1–6. doi: 10.1109/QPAIN69676.2026.11546620.

- Curated 2,845 protein sequences from nine native Bangladeshi fish species and benchmarked seven architectures; MotifCNN-Transformer+TA-PE achieved 79.80% accuracy with GPU-free inference while running 5× faster and 42× smaller than ProtBERT.

Manuscripts Under Consideration

Parameter-Efficient Preference Optimization and Supervised Fine Tuning of ProGen2 for Thermophile Sequence Space

Md Nasiat Hasan Fahim, Md. Abid Ullah Muhib, and Mohammad Shahidur Rahman. *PLOS ONE*. **Submitted; 2026.**

- Adapted ProGen2 toward thermophile-associated protein sequences using LoRA-based SFT, DPO, and KTO, revealing distinct likelihood-preference trade-offs under controlled compute and homology-controlled evaluation.

Spatiotemporal Crime Dynamics and Recidivism Analysis in Urban Bangladesh: A Hybrid K-Means Approach

Ahsan Habib, Md. Mehedi Hasan, Rayhanul Amin Tanvir, Abu Sayeid Sawon, **Md Nasiat Hasan Fahim**, and Nixon Deb Antu. *SUST Journal of Science and Technology*. **Submitted, 2026.**

- Analyzed Sylhet crime patterns using newspaper and police data, combining spatiotemporal clustering, recidivism analysis, and forecasting to identify hotspots, predict future crime trends, and derive public-safety insights.

Active Research

PROTEIN LANGUAGE MODEL

- ‘De-average’ evolutionary signals in protein language models from universal average-protein translators to lifestyle-specific models. This research utilizes SOTA finetuning approaches and Explainable AI to recover biophysical adaptation rules for extremophiles.

SAR SEMANTIC SEGMENTATION

- Developing cross-regional Sentinel-1 water-segmentation models while investigating geographic and biome bias, backscatter variability, radar noise and shadow effects, and model behavior through systematic evaluation and explainable AI.

EARTHQUAKE EARLY WARNING

- Developing globally generalizable early-warning models from initial P-waves, linking waveform characteristics with earthquake shaking, energy, and source information while studying cross-region generalization and model behavior using explainable AI.

Competitive Programming

Codeforces

SPECIALIST · MAX RATING: 1431 · HANDLES: MASTEROFTHESEA, ENBUGGER

CodeChef

3 STAR CODER · MAX RATING: 1735 · HANDLE: MASTEROFTHESEA

Codeforces, CodeChef, LightOJ, AtCoder, Leetcode

1500+ PROBLEMS SOLVED ACROSS PLATFORMS

COMPETED IN 2 NATIONAL INTER-UNIVERSITY PROGRAMMING ONSITE CONTESTS

Projects

Multimodal AI Search & Recommendation Engine



FASTAPI · OPENAI VISION · CHROMADB · REDIS · POSTGRESQL · DOCKER

- Full-stack e-commerce platform featuring an AI-powered search and recommendation system. Multimodal search microservice (FastAPI) utilizing OpenAI Vision and ChromaDB for hybrid text/image retrieval, Redis caching and a robust PostgreSQL-to-vector synchronization pipeline deployed in Docker.

COWTRAIN: Cattle Weight Estimation

YOLOV8 · INSTANCE SEGMENTATION · REGRESSION · FASTAPI

- End-to-end ML product for weight prediction from images. A computer vision pipeline utilizing YOLOv8 for instance segmentation of cow and the reference object and regression to accurately predict weight, FastAPI backend.

Achievements

Champion, AI MEETS CLIMATE Hackathon

Oxfam

Real-time climate-reporting pipeline with ASR and NER models

Awarded, SUST Rising Team

SUST

Best-performing junior team in CSE Carnival 2024

2nd Runner Up, AI-FICATION 2025

CUET

ASR for converting 10 regional languages into standard Bengali

3rd Place, Robi Datathon 3.0 Qualifier — Problem 3

Robi

Future purchase-probability forecasting with time-series analysis

4th Place, DL Enigma 1.0: SUST CSE Carnival

SUST

Bangladeshi roadside object detection

5th Place, Code Samurai Hackathon 2024

DU and BJIT

Optimal garbage-collection routing and service planning

Skills

Languages

C, C++, Python, Java

AI & Data Science

Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, PyTorch, TensorFlow, LangChain

Backend & Database

FastAPI, SQL, PostgreSQL, RAG, Vector Database, AWS Services, Docker, Git

Certifications

IBM Data Science Professional Certificate

IBM (Coursera)

DATA SCIENCE LIFECYCLE AND GENERATIVE AI

Machine Learning Specialization

Stanford University (Coursera)

SUPERVISED, UNSUPERVISED AND REINFORCEMENT LEARNING

AWS Certified Solutions Architect - Associate

AWS Training and Certification

CERTIFIED SOLUTIONS ARCHITECT - ASSOCIATE (SAA-C03)

Cloud Foundations

AWS Academy

CORE CLOUD COMPUTING CONCEPTS

Cloud Developing

AWS Academy

TECHNICAL EXPERTISE IN DEVELOPMENT WITH CLOUD TECHNOLOGIES

Referees

Mohammad Shahidur Rahman, PhD

Professor
Department of CSE, SUST

Undergraduate Thesis Supervisor
& Course Instructor

rahmanms@sust.edu

Dr. Ahsan Habib

Associate Professor
IICT, SUST

Research Assistant Supervisor

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Dr. Md Forhad Rabbi, SMIEEE

Professor & Head
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Research Collaborator
& Course Instructor

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